Sections: B,C & D

Linear Algebra (MT-1004)

Homework#36

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Question 1

Quadratic Form and Eigenvalues

Let

$$A = \begin{bmatrix} 4 & 2 \\ 2 & 3 \end{bmatrix}$$

(a) Represent the quadratic form $Q(x_1, x_2) = x^T Ax$.

(b) Find the eigenvalues of A and use them to determine the definiteness of $Q(x_1, x_2)$.

Charcterization of Positive Definite

Consider the matrix

$$A = \begin{bmatrix} 6 & -2 & 1 \\ -2 & 5 & -1 \\ 1 & -1 & 4 \end{bmatrix}$$

(a) Perform the **leading principal minors test** to determine if A is positive definite.

(b) Verify your result by checking if all eigenvalues of A are positive.

Question 2

Positive Definite Matrix in Optimization

An optimization problem involves minimizing the quadratic form:

$$f(x_1, x_2) = 2x_1^2 + 3x_2^2 + 4x_1x_2.$$

(a) Express $f(x_1, x_2)$ in matrix form as $f(x) = x^T Ax$.

(b) Check if A is positive definite.

(c) Explain the role of positive definiteness in ensuring the existence of a unique minimum for $f(x)$.